GastroOps AI — Complete System Architecture

End-to-end workflow: Yelp data ingestion → data layer → hybrid retrieval → multi-agent analysis → manager report

START — Yelp Open Dataset
Business records • Customer reviews • Ratings • Categories • Locations • Dates

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Data Ingestion & Preprocessing
Load records • Clean and prepare review text • Process batches • Prepare metadata

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PostgreSQL + pgvector (PROPOSED CONSOLIDATED DATA LAYER)
Business records • Reviews • Vector embeddings • Feedback • Searchable text

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Embedding Generation & Search Indexing
Generate review embeddings • Prepare vector similarity and keyword search

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USER QUERY PROCESSING

Restaurant Manager
Enters a natural-language question

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React Frontend → FastAPI Backend
Submit query • API routing • Input validation guardrails

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Query Understanding & Embedding Generation
Interpret the question • Generate a query vector for semantic search

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Keyword Search
Full-text search or BM25-compatible scoring

Semantic Search
pgvector similarity search

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Hybrid Retrieval
Combine results • Rank fusion • Reranking • Metadata filtering

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Evidence Validation
Check relevance • Validate supporting reviews • Filter weak evidence

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Multi-Agent Orchestration
Coordinate complaint analysis (label LangGraph only if used in the implementation)

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Complaint Classification & Prioritization
Classify complaint types • Assign severity and priority

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Aspect & Root-Cause Analysis
Identify recurring issues and evidence-supported possible causes

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Operations Strategist Agent
Recommend corrective actions and standard operating procedures (SOPs)

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LLM Provider Layer
Configured AI providers • Provider fallback when available

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Quality Evaluation & LLM-as-Judge
Evaluate answer quality, relevance and evidence grounding

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Executive Summary Generation
Synthesize findings, supporting evidence and recommendations

SUPPORTING SERVICES (RUN WHEN REQUESTED / CONFIGURED)

CSAT Prediction
TF-IDF + Ridge Regression

Complaint Trend Prediction
Historical complaint patterns

Summary Translation
English • Kannada • Hindi • Telugu • Tamil

Manager Feedback
Ratings • Feedback • Quality improvement

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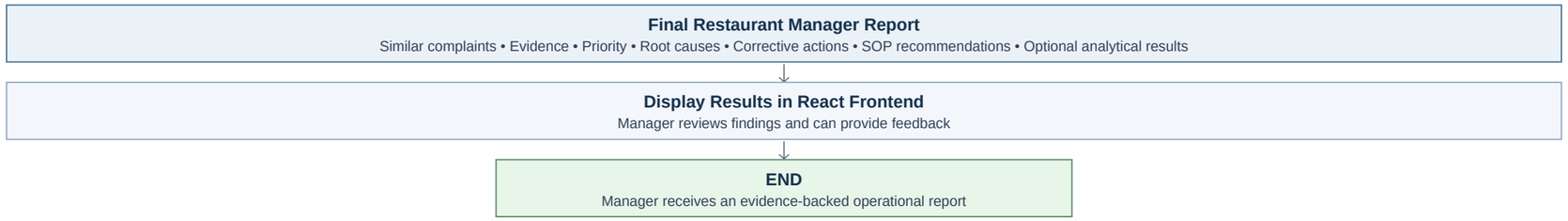

Design note: PostgreSQL + pgvector is a proposed consolidated design, not the current implementation. Full-text search is not automatically BM25. Label LangGraph as implemented only if the code actually uses it. Supporting services may run independently rather than on every query.